

Data

DATASET STRUCTURE

Training Set : 300 homes
Testing Set : 200 homes

TARGET VARIABLES

- 1. Sale Price (Continuous, kEUR)
- 2. Distressed Status (Binary 0/1)

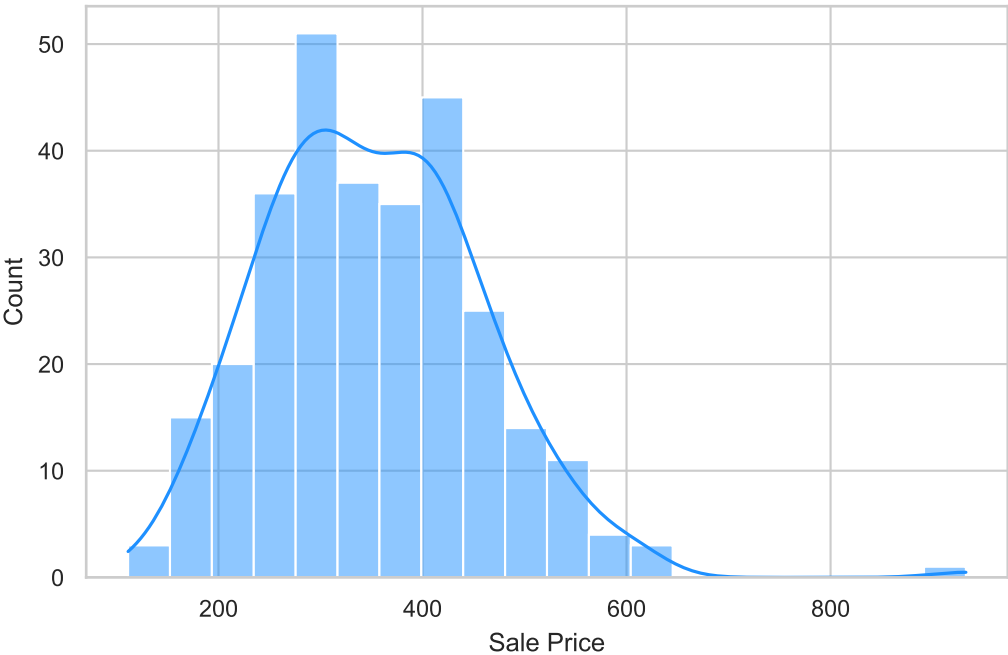
PREDICTORS (12 Features)

GrLivArea, LotArea, OverallQual, OverallCond, YearBuilt, YearRemodAdd, TotalBsmtSF, GarageArea, BedroomAbvGr, Fireplaces, WoodDeckSF, OpenPorchSF

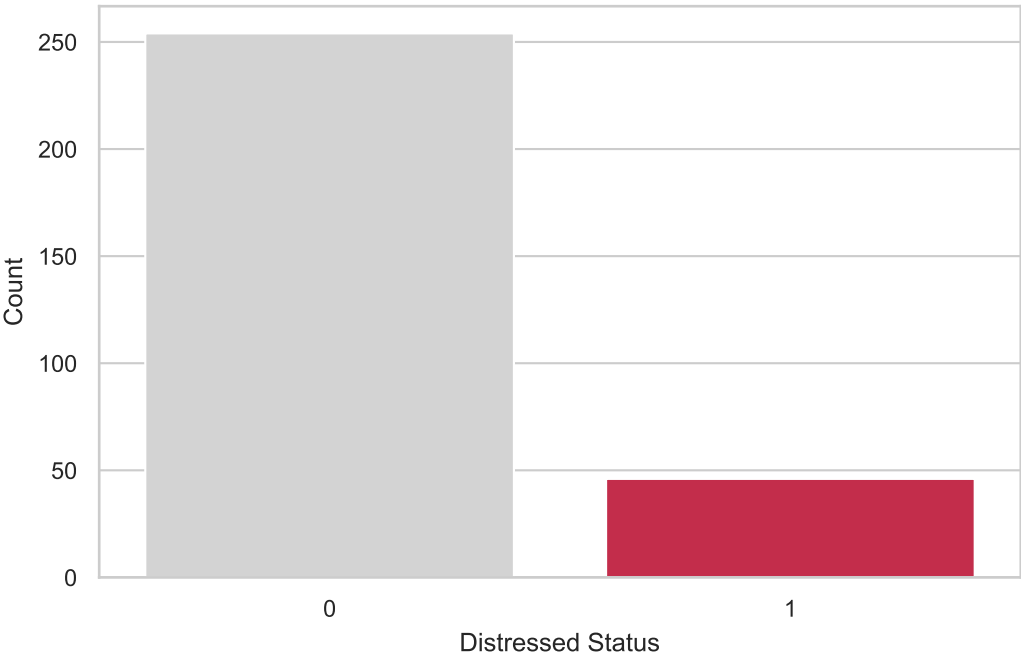
SUMMARY STATISTICS (KEY FEATURES)

	sale_price_keur	GrLivArea	OverallQual	YearBuilt
count	300.0	300.0	300.0	300.0
mean	352.6	1524.3	6.1	1971.2
std	106.9	545.4	1.4	31.8
min	111.4	480.0	2.0	1880.0
25%	276.6	1144.8	5.0	1950.0
50%	345.9	1460.0	6.0	1973.0
75%	420.2	1768.0	7.0	2003.0
max	932.7	4676.0	10.0	2009.0

Target 1: Distribution of Sale Prices (kEUR)



Target 2: Distressed Class Imbalance (15.3%)



Model, Method and Results

Model Formulas

1. Random Forest Regression / Classification Architecture:

$$f_{RF}(x) = \frac{1}{B} \sum_{b=1}^B T_b(x), \text{ where } m_{try} = 4 \text{ randomly drawn}$$

2. Gradient Boosting Optimization Equation:

$$F_M(x) = \sum_{m=1}^M \eta h_m(x), \text{ where } \eta = 0.05$$

Model Assumptions

- Sampling Independence: Observations are assumed I.I.D. Drawn from stable population.
- Invariant Split Monotonicity: Splits depend on ordinal ranks. Completely immune to outliers.
- Imbalance Calibration: Assumes minority class requires synthetic weights for cross-entropy.

Methodology & Libraries Used

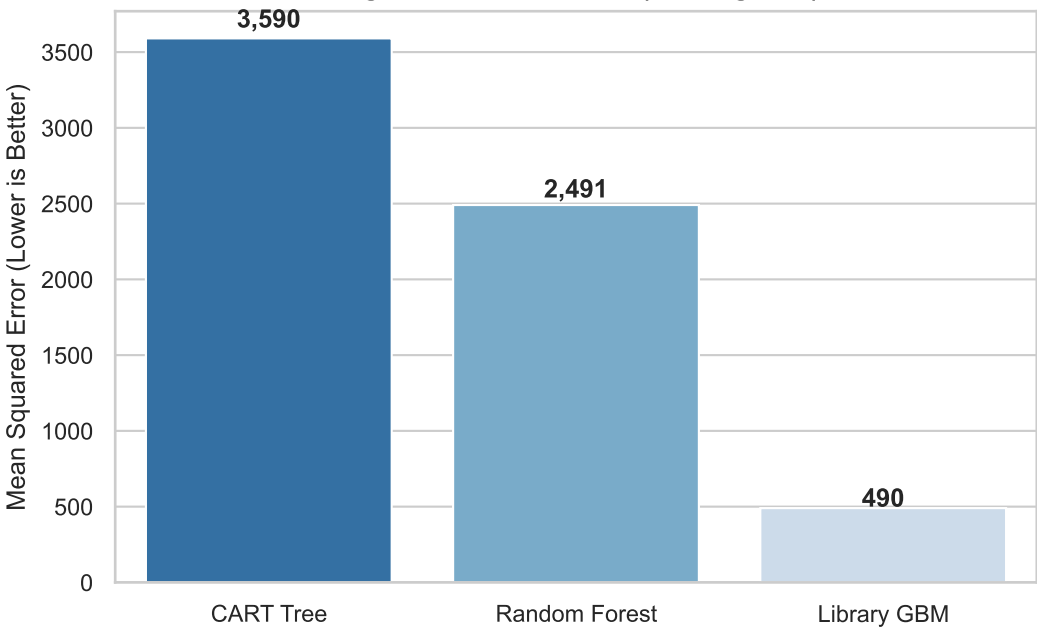
- NumPy: Core CART Trees built strictly from scratch.
Custom MDI Vector Logic: Programmed mathematical SSE tracking.
- Scikit-Learn: Executed benchmark architectures (GBM).
Probability Calibration: Deployed balanced weights to handle 15% class prevalence.

Model Results

- Task 1: Single CART Regressor (From Scratch)
Target: sale_price_keur
Training MSE: 3,590.46
- Task 2: Random Forest Regressor (From Scratch)
Target: sale_price_keur
Training MSE: 2,490.72
- Task 3: Gradient-Boosted Regressor (Library)
Target: sale_price_keur
Training MSE: 490.28 (Scikit-Learn)
- Task 4: Random Forest Classifier (Library)
Target: distressed state
Log-Loss: 0.2642 (Balanced Weights)
- Task 5: Gradient-Boosted Classifier (Library)
Target: distressed state
Log-Loss: 0.2856 (Sample Weights)

Slide 3: Performance Comparisons & Architecture Visualization

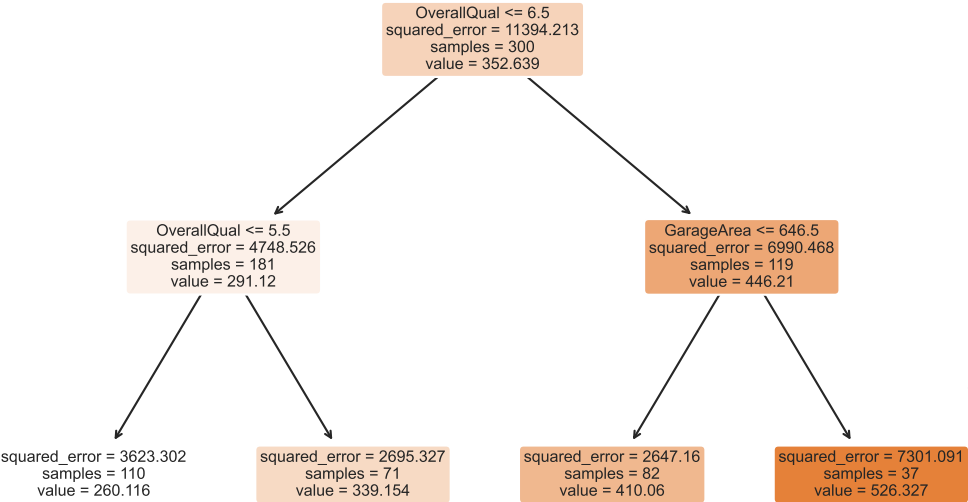
Regression Performance (Training MSE)



Classification Performance (Cross-Entropy Loss)



CART Architecture (Extract of Top Nodes)



Top 5 Drivers of Price (MDI Importances)

